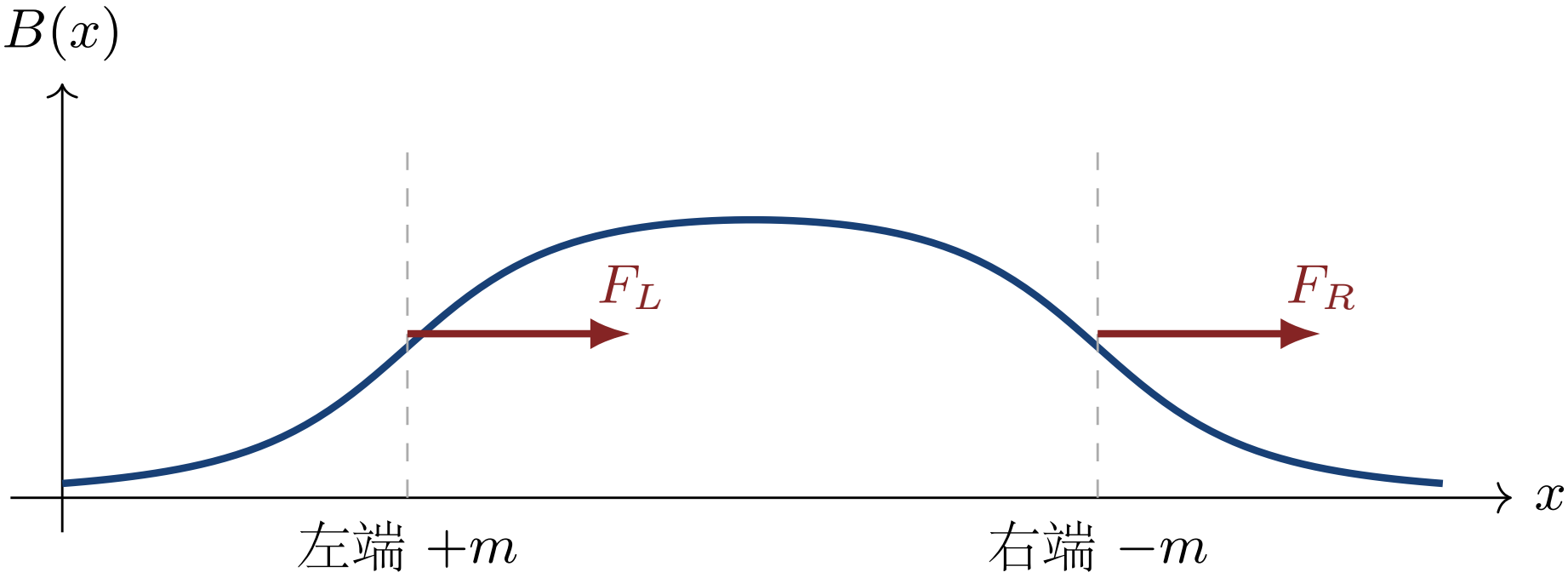


两端力同向：$F = \mathcal{A}I$

$$\varepsilon = IR + \mathcal{A}v,\ \ kv = \mathcal{A}I$$

$$v = \mathcal{A}\varepsilon/(kR + \mathcal{A}^2)$$